



RESEARCH ARTICLE

Safe and Privacy-Aware AI Models for Medical Image Processing: A Multi-Paradigm Comparative Study

Shivam Tiwari^{1*}, Mukta Bhatele², Akhilesh A. Wao³

^{1,2,3} Department of Computer Science, AKS University, Satna, Madhya Pradesh, India – 485001

*Email: shivamtiwari52@yahoo.com

Email: 30.muktabhatele@gmail.com

Email: akhileshwao@gmail.com

*Corresponding author

Received: 02 July 2026 | Revised: 18 August 2026 | Accepted: 08 September 2026 | Published: 26 September 2026

DOI: 10.5281/zenodo.22975795

How to cite this article: Shivam Tiwari, Mukta Bhatele, Akhilesh A. Wao (2026). Safe and Privacy-Aware AI Models for Medical Image Processing: A Multi-Paradigm Comparative Study. International Journal of Advanced Computing and Machine Intelligence (IJACMI), 1(1), pp. 44-54. DOI: 10.5281/zenodo.22975795

Abstract

In addition to producing better models, hospitals that combine patient scans into a single training server also provide a single, alluring target: in 2023 alone, over 130 million healthcare records were compromised globally. This research begins with this contradiction between the confidentiality requirements of medicine and the data hungry of deep learning. We maintain the same architecture, preprocessing, hyperparameters, and evaluation protocol while training identical ResNet-50 classifiers on the NIH Chest X-ray corpus and a BraTS brain-tumor MRI set under three privacy regimes: Federated Learning, Differential Privacy, and Homomorphic Encryption. Federated Learning loses just 0.7 points (92.4%) while never sending raw images off-site, whereas centralized training achieves 93.1% accuracy. At $\epsilon = 1.0$, Differential Privacy significantly reduces membership-inference attacks at a higher cost (89.7% accuracy). At the expense of about four times slower inference, homomorphic encryption maintains correctness at 92.8% and provides the highest confidentiality guarantee of the three. In bandwidth-tolerant multi-hospital contexts, Federated Learning provides the optimal accuracy-for-privacy exchange, while Differential Privacy and Homomorphic Encryption are better suited to institutions with lower latency budgets and stricter regulatory or secrecy limitations.

Keywords: Federated Learning · Differential Privacy · Homomorphic Encryption · Medical Image Processing · Secure DICOM · Healthcare Cyber-Security

1.0 Introduction

These days, deep convolutional networks can interpret chest X-rays and identify brain cancers with accuracy comparable to that of skilled radiologists [1], [4]. In circumstances where minutes matter, this skill has swiftly transitioned from research papers into clinical pipelines, reducing the time between a scan and a diagnosis. The catch is how these models get trained. Building a CNN that generalizes well often entails pooling thousands of scans — often DICOM files including identifying patient metadata — onto one server or cloud platform. Centralizing that much sensitive imaging data transforms the training server become a high-value target, and healthcare has routinely rated among the most attacked sectors for just this reason [10]. Adversaries with merely query access to a deployed model may occasionally deduce if a certain patient's scan was included in the training set or partially recreate training images from the model's parameters, making a model trained in this manner vulnerable to direct attack. Three engineering approaches to this problem — Federated Learning, Differential Privacy, and Homomorphic Encryption — are the subject of this study.

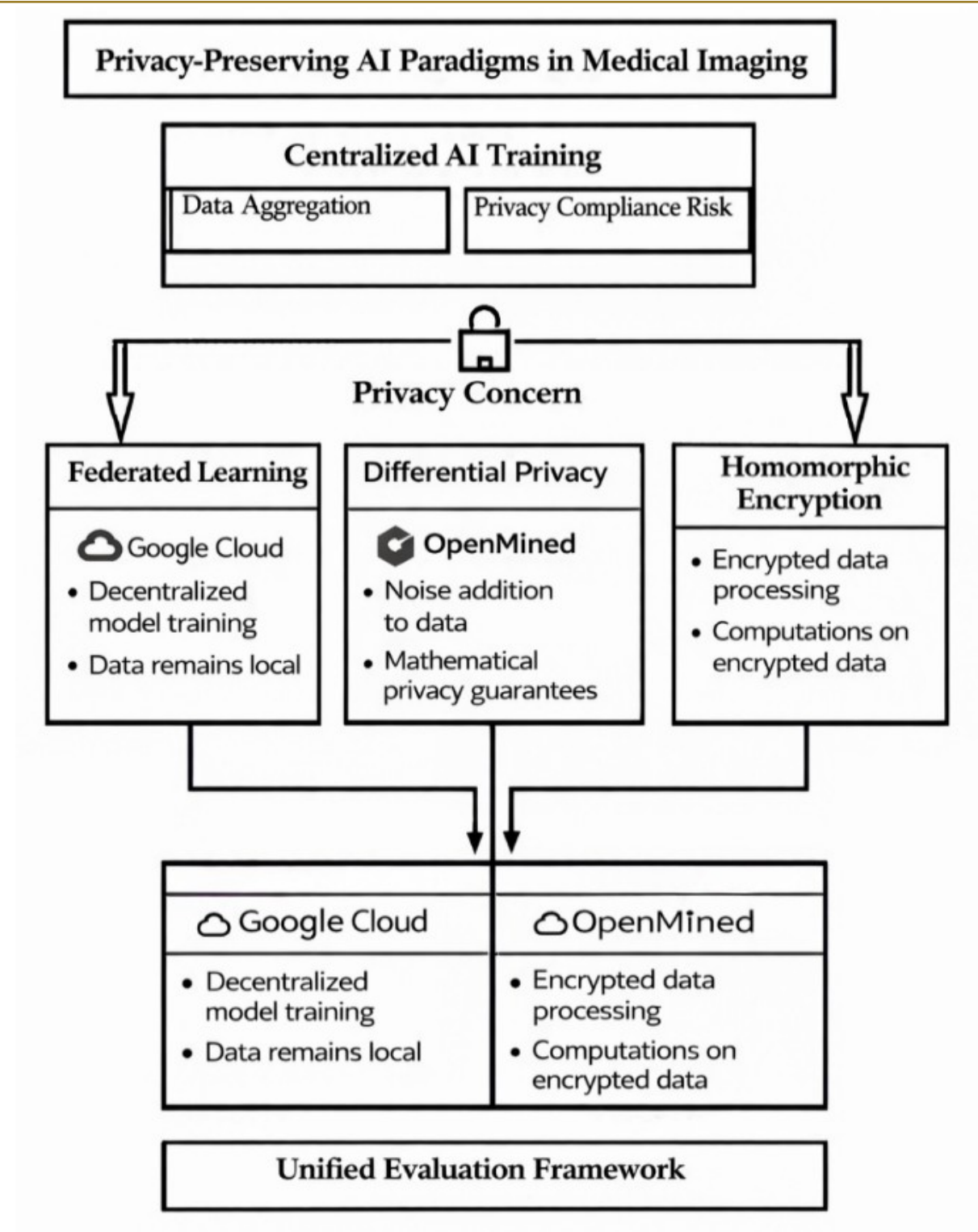


Fig. 1. Privacy-Preserving AI Paradigms in Medical Imaging.

Each paradigm steps in at a distinct stage of the process, as shown in Figure 1. Federated Learning never moves the raw scan at all – only model changes leave the hospital. Differential privacy allows the data to migrate (or remain local) while introducing calibrated noise into the training signal to prevent the isolation of a particular patient's contribution later on. The reverse strategy is used in homomorphic encryption, where computation is done directly on the ciphertext and the data can be encrypted before it even leaves the hospital. Each purchases privacy at a different price, which is exactly what this paper's remaining sections quantify.

1.1 Rapid Adoption of AI in Radiology

Compared to most other clinical disciplines, radiology adopted deep learning earlier and more quickly. In published benchmarks, chest X-ray classifiers can already diagnose thoracic diseases with 90% accuracy [4], and multi-site federated pilots have demonstrated that the same architectures can be trained collaboratively across hospitals without a central data lake [5]. The flipside of that achievement is dependency: as triage systems lean more heavily on AI output, the underlying infrastructure — usually a centralized cloud environment chosen for its scalability — becomes a bigger problem from a confidentiality and compliance standpoint.

1.2 Convolutional Neural Networks in Medical Imaging

Current medical imaging research is dominated by ResNet and DenseNet-style architectures because its skip connections allow very deep networks to be trained without the gradient signal disappearing. Large labeled datasets are still needed for effective training, and these datasets are often kept in one location. Membership-inference and model-inversion attacks, which let an adversary to recover traces of the training data from a model that was never intended to expose it, precisely take advantage of this concentration [10]. Any option that protects privacy must be able to withstand these particular attacks, not merely sound safe in theory.

1.3 Privacy Risks in Centralized AI Training

A deep network trained without any privacy measure can memorize and leak individual training examples, as demonstrated years ago by Abadi et al. [6]. In addition to the technological risk, centralized pipelines also have a legal component: HIPAA and GDPR both place stringent, auditable regulations on the storage, transfer, and processing of patient data, with actual financial consequences for non-compliance [10]. Therefore, selecting a privacy-preserving training approach is more of a compliance obligation for many institutions than an optional hardening.

1.4 Privacy-Preserving AI Paradigms

Federated Learning (FL) maintains raw data at each institution and only distributes model parameters; secure aggregation procedures are introduced to prevent information leakage from the parameter exchange itself [1]–[3]. To ensure that no patient's record can be reliably identified in the output, Differential Privacy (DP) injects calibrated noise into training. The strength of this assurance is determined by a privacy budget [6]. Homomorphic Encryption (HE) computes directly on encrypted data, so the model never sees plaintext patient information even during inference — at the penalty of much costlier computation [11], [12].

Each of these has been investigated independently. What is missing is a direct, apples-to-apples comparison of all three under one dataset, one architecture, and one assessment process — which is the gap this paper tackles.

1.5 Research Gap and Motivation

The majority of published assessments select a single paradigm and benchmark it separately [3], [13]. That makes it impossible for a hospital IT staff to answer a very practical question: given our accuracy floor, our latency budget, and our regulatory risk, which of these three should we actually deploy? Answering that requires putting FL, DP, and HE through the identical dataset, architecture, and metrics side by side — which is what Sections 3–5 of this study do.

1.6 Contributions of This Study

This paper makes the following contributions: (i) a shared benchmarking pipeline that trains identical CNNs under FL, DP, and HE; (ii) comparisons of accuracy, F1, and AUC against a centralized baseline on two publicly available medical imaging datasets; (iii) an empirical assessment of each paradigm's resistance to membership-inference attacks; (iv) a measured computational-overhead comparison (training time, communication cost, and inference latency) as opposed to a theoretical one; and (v) deployment recommendations that link each paradigm to the institutional context in which it works best.

2.0 Related Work

This section examines previous research in five domains that directly affect the comparison in Section 5: the three privacy models examined here, general AI applications in medical imaging, and what comparative work has already been done.

2.1 AI in Medical Image Processing

Chest X-ray triage, brain-tumor segmentation, retinal disease screening, and histological analysis are now handled by CNN-based classifiers at a level that ten years ago would have required a subspecialist. In particular, Sheller et al. [4] demonstrated that distributed training can match centralized accuracy on medical pictures, and Dayan et al. [5] expanded on that by forecasting COVID-19 outcomes across 20 hospitals without combining their data. Many of these findings are based on the NIH Chest X-ray corpus, which has over 100,000 pictures and often pushes accuracy above 90%. However, almost all of it was trained in a centralized manner, and Hu et al. [10] specifically showed that models built in this manner can reveal whether or not a particular patient's scan was used during training.

2.2 Federated Learning in Healthcare

FL was first introduced by McMahan et al. [1] as a collaborative training method without centralizing data, then Bonawitz et al. [2] used secure aggregation to fix a clear flaw in the naive version: the parameter updates themselves. The factors that still make FL challenging in practice were listed by Kairouz et al. [3]: non-IID data across clients, communication overhead, and convergence instability, all of which are present in actual hospital deployments. Sheller et al. [4] and Dayan et al. [5] both ran FL on real clinical imaging tasks and reported it works, but neither paper claims FL is leak-proof — gradient updates can still carry information without secure aggregation, and communication cost remains a real constraint for hospitals on constrained networks.

2.3 Differential Privacy in Deep Learning

The reference technique used here is Abadi et al.'s DP-SGD [6], which noises and clips gradients during training so that the final model has a provable (ϵ, δ) privacy guarantee. With PATE, Papernot et al. [9] adopted a different approach, educating a group of instructor models and privately imparting their expertise to a pupil. Both ways work; both also cost accuracy, and the cost increases steeper as ϵ lowers. Hu et al. [13] found this degradation especially obvious on high-dimensional medical images, which makes the choice of ϵ in Section 5.2 a genuinely relevant design decision rather than a default value.

2.4 Homomorphic Encryption for Secure AI

CryptoNets [12] was the demonstration that neural-network inference on encrypted data is conceivable at all, and Wood et al. [11] assessed how that idea has advanced toward healthcare especially. The trade-off is inevitable: as HE schemes only handle addition and multiplication natively, nonlinear activations must be estimated and plaintext arithmetic is replaced with significantly heavier encrypted arithmetic, which is usually 3–5× slower at inference. The FL and DP literature does not have to deal with the accuracy loss caused by that approximation stage.

2.5 Comparative Studies and Identified Limitations

What's largely absent from this literature is a controlled, three-way comparison. Studies either benchmark one paradigm against a centralized baseline [3], [13], or survey the field theoretically without new experiments. Attack resistance in particular is rarely measured alongside accuracy in the same paper — a study will report DP's theoretical guarantee, or FL's accuracy, but seldom both against a shared threat model. Table 1 summarizes what each cited study actually measured, and the gaps are visible column by column.

Table 1. Comparative Summary of Existing Studies

Study	Privacy Method	Dataset	Accuracy	Limitation
McMahan et al. [1]	Federated Learning	MNIST (non-medical)	~99%	Not evaluated in medical imaging
Sheller et al. [4]	Federated Learning	Brain Tumor MRI	Comparable to centralized	Limited attack-resistance analysis
Dayan et al. [5]	Federated Learning	Multi-hospital COVID CT/X-ray	>90%	High communication overhead
Abadi et al. [6]	Differential Privacy (DP-SGD)	MNIST / CIFAR	Reduced accuracy at low ϵ	Performance degradation under strict privacy
Papernot et al. [9]	Differential Privacy	MNIST	Competitive accuracy	Scalability limitations

	(PATE)			
CryptoNets [12]	Homomorphic Encryption	MNIST	98%	3–5× computational latency
Wood et al. [11]	HE Survey	Healthcare datasets	Theoretical analysis	High computational complexity
Hu et al. [10]	Attack Analysis	Various ML models	N/A	Demonstrates vulnerability of centralized models

None of these evaluate FL, DP, and HE together on the same medical dataset with the same attack model — that gap is what Sections 3–5 close.

3.0 Methodology

To make the three paradigms comparable at all, everything except the privacy mechanism itself was held fixed: same datasets, same preprocessing, same CNN backbone, same evaluation metrics. Sections 3.1–3.4 walk through that pipeline in the order it was actually built — datasets first, then the baseline model, then the comparative framework.

3.1 Dataset Description

Two public datasets were used: the NIH Chest X-ray dataset for pneumonia detection, and BraTS (Brain Tumor Segmentation) MRI for tumor classification. Both are standard benchmarks in this literature, which makes results here easier to compare against prior single-paradigm studies.

Table 2. Summary of Medical Imaging Datasets Used for Evaluation

Dataset	Modality	Volume / Images	2D Images Used	Target Task
NIH Chest X-ray	X-ray (frontal view)	112,120 images	25,000 (balanced subset)	Pneumonia detection (binary classification)
BraTS MRI	MRI (T1, T1Gd, T2, FLAIR)	~2,000 volumes	>100,000 slices	Tumor presence classification

The 25,000-image NIH subset was drawn to give a roughly balanced positive/negative split for pneumonia detection, since the full 112,120-image corpus is heavily skewed toward negative cases.

3.1.1 Preprocessing Steps. Every image was resized to 224×224 and pixel intensities normalized to [0,1] to match the ResNet-50 input and stabilize training. DICOM header metadata was stripped before any image entered the pipeline, since headers routinely carry patient identifiers that have nothing to do with the image content itself. Oversampling corrected the pneumonia class imbalance, and BraTS volumes were sliced into 2D axial images while keeping tumor regions intact across slices.

3.1.2 Data Augmentation. Training images (only training images — never validation or test) were augmented with horizontal flips, small rotations ($\pm 15^\circ$), brightness/contrast jitter, $\pm 10\%$ zoom, and Gaussian noise. Two adjustments were paradigm-specific: augmentation strength under DP was kept conservative to avoid stacking augmentation noise on top of the privacy noise already being injected, and under FL each simulated client applied augmentation independently, which better mimics how real hospital data actually varies from site to site.

3.2 Baseline Model

A single baseline model anchors every comparison in Section 5. Its job is to establish what accuracy is achievable with no privacy constraint at all, so that every privacy-preserving result can be read as a cost relative to that ceiling rather than in isolation.

3.2.1 Network Architecture. ResNet-50 was the primary architecture, chosen for its track record on medical imaging tasks and its skip connections, which keep gradients from vanishing at this depth. The final fully connected layer was swapped for binary classification. DenseNet-121 was run as a secondary check on the same

data — results tracked ResNet-50 closely, so ResNet-50 is what's reported throughout. All models used batch size 32, learning rate 0.001, 50 epochs, a cosine learning-rate schedule, and early stopping at patience 10.

Binary Cross-Entropy loss (Eq. 1) was the objective for both classification tasks:

$$L = -(1/N) \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

Where y_i represents the ground truth label and $\hat{y}_i$ represents the predicted probability. Binary cross-entropy was selected due to its probabilistic interpretability and stable gradient properties. In Differential Privacy experiments, the loss gradients were clipped and noised per the DP-SGD procedure before the optimizer step. Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$) handled optimization throughout.

3.3 Proposed Comparative Framework

Figure 2 shows how the three paradigms plug into one evaluation harness rather than three separate ones. The point of building it this way was to make sure any difference in the results traces back to the privacy mechanism itself, not to some incidental difference in how each paradigm happened to be implemented.

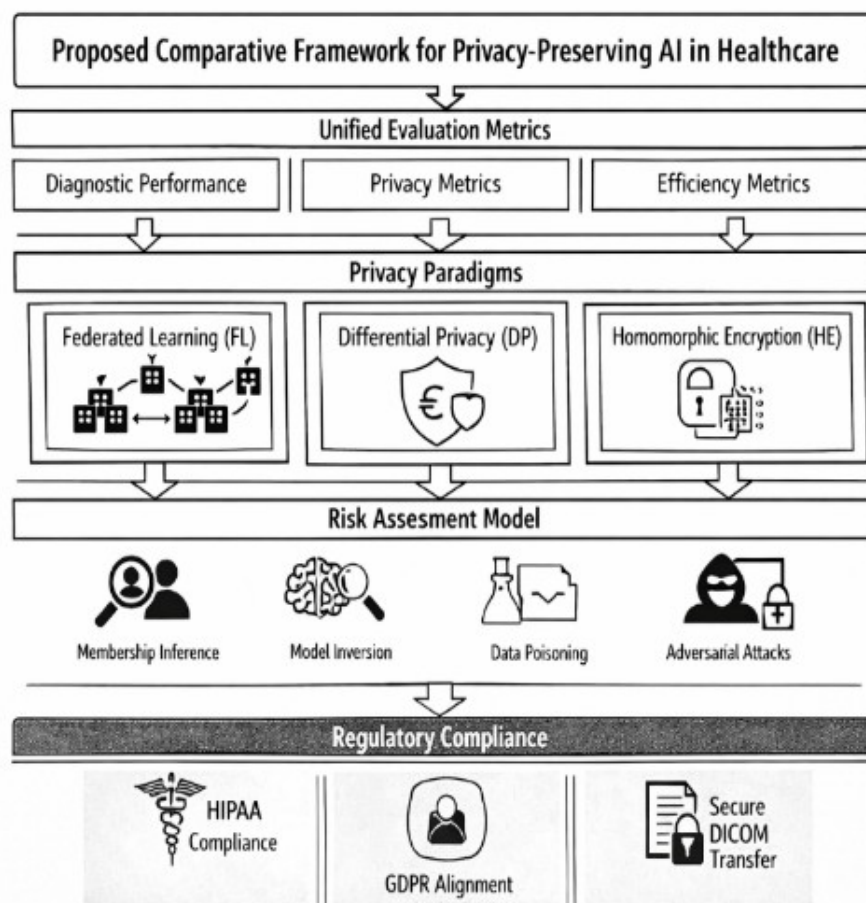


Fig. 2. Proposed Comparative Framework for Privacy-Preserving AI in Medical Image Processing.

The framework tracks three metric families in parallel — diagnostic performance (accuracy, F1, AUC), privacy strength (ϵ , resistance to gradient leakage), and efficiency (training time, communication overhead, inference latency) — against a shared risk model covering membership inference, model inversion, data poisoning, and adversarial perturbation, and against HIPAA/GDPR/DICOM compliance requirements.

3.3.1 Unified Training Pipeline. The centralized model serves as reference. FL distributes the same dataset across simulated clients; DP and HE are applied within the identical CNN configuration used for the centralized run. Every model passes through the same five stages: preprocessing and augmentation, initialization, privacy-mechanism integration, training and validation, then held-out test evaluation. Keeping this sequence identical across all four conditions is what makes the Section 5 comparison meaningful rather than incidental.

3.4 Standard Evaluation Metrics

Table 3. Categories of Evaluation Metrics Used

Category	Metric(s)
Performance	Accuracy, F1-score, AUC
Privacy	ϵ -value (DP), Encryption strength (HE)
Efficiency	Training time, Communication cost (FL), Inference latency (HE)
Security	Resistance to membership- and model-inference attacks

These four categories map directly onto the four questions a deploying hospital actually asks: how accurate, how private, how expensive, and how attack-resistant.

4.0 Experimental Setup

All four conditions — centralized, FL, DP, HE — ran on identical hardware and software, back to back, to remove infrastructure as a confound.

4.1 Hardware Configuration

CPU: Intel Core i9 (10 cores); GPU: NVIDIA RTX 3080 (10 GB VRAM); RAM: 32 GB; Storage: 1 TB SSD; Operating System: Ubuntu 22.04.

4.2 Software Libraries and Training Configuration

PyTorch 2.0 ran the core training loop; TensorFlow Privacy handled the DP mechanism; PySyft simulated the federated clients; and NumPy, Pandas, and Scikit-learn covered data handling and metrics.

Epochs: 50; Batch size: 32; Learning rate: 0.001; Communication rounds (FL): 100; Local epochs per client (FL): 2.

5.0 Results

Sections 5.1–5.3 report accuracy, privacy strength, and computational cost for all four conditions, evaluated on identical held-out test data.

5.1 Diagnostic Accuracy Comparison

Federated Learning came closest to matching the centralized baseline, giving up under a full point of accuracy. Differential Privacy showed the clearest cost curve — accuracy dropped as ϵ tightened, exactly as the theory predicts. Homomorphic Encryption held accuracy nearly at baseline, which makes sense given it changes inference, not training.

Table 4. Diagnostic Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-Score	AUC
Centralized (Baseline)	93.1	0.92	0.91	0.92	0.95
Federated Learning (FL)	92.4	0.91	0.90	0.91	0.94
Differential Privacy ($\epsilon = 1.0$)	89.7	0.88	0.87	0.88	0.91
Differential Privacy ($\epsilon = 0.5$)	87.9	0.86	0.85	0.86	0.89
Homomorphic Encryption (inference)	92.8	0.91	0.90	0.91	0.94

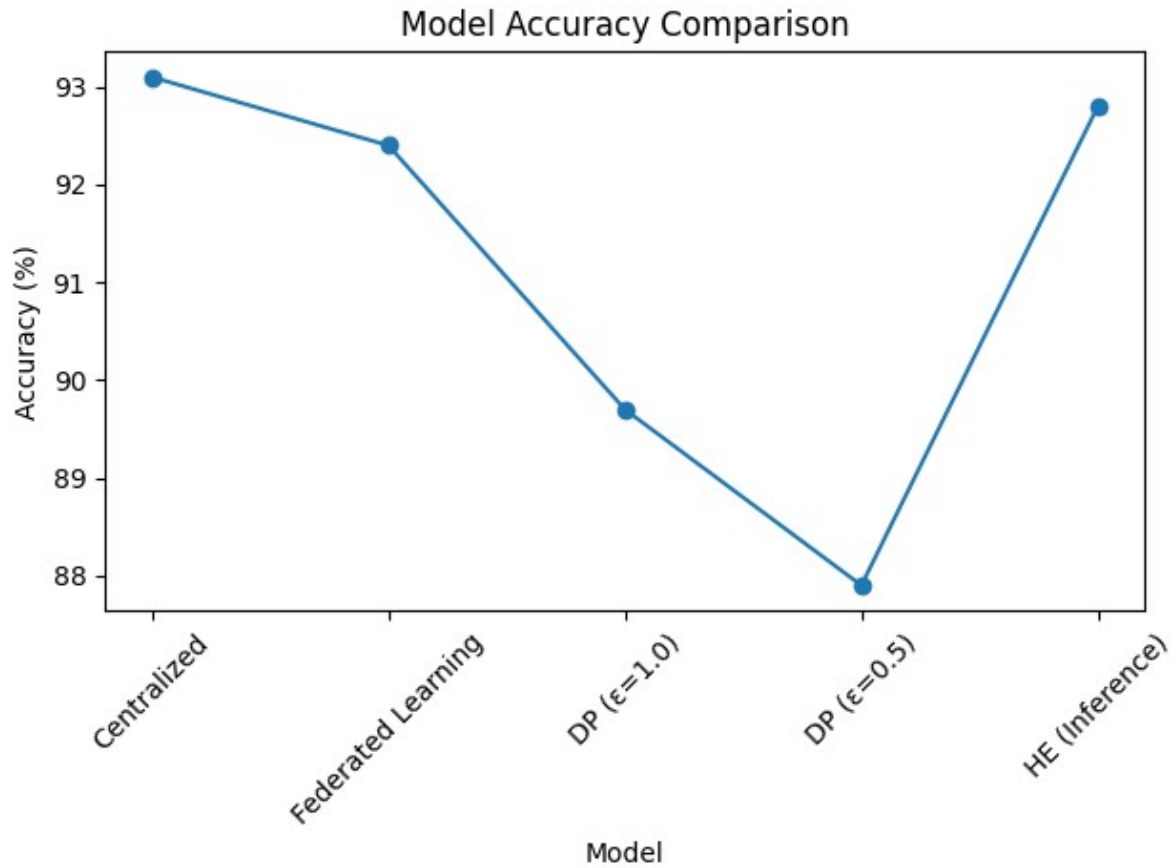


Fig. 3. Diagnostic Accuracy Comparison of Privacy-Preserving AI Models.

Tightening ϵ from 1.0 to 0.5 cost 1.8 accuracy points — a steeper penalty than moving from centralized training to full Federated Learning. That gap is the clearest single number in this study for anyone deciding between FL and DP on accuracy grounds alone.

5.2 Privacy Strength Analysis

Accuracy is only half the picture; Table 5 scores the same four conditions on privacy strength and resistance to membership-inference attacks.

Table 5. Privacy Strength Evaluation

Method	Privacy Level	Attack Resistance
Centralized	Low	High vulnerability
Federated Learning	Moderate	Reduced exposure (gradient-leakage risk)
Differential Privacy ($\epsilon = 1.0$)	High	Strong protection
Differential Privacy ($\epsilon = 0.5$)	Very High	Very strong protection
Homomorphic Encryption	Very High	Maximum protection

Put the two tables side by side and the trade-off is stark: the two strongest privacy scores (DP at $\epsilon = 0.5$, and HE) both come with a real cost elsewhere — DP in accuracy, HE in latency, which Section 5.3 quantifies next. Federated Learning is the only condition that doesn't pay for its privacy gain in either of those currencies, though it isn't the strongest privacy guarantee of the three either.

5.3 Computational Overhead

Privacy protection is not free computationally, and the size of that bill differs sharply by paradigm.

Table 6. Computational Cost Comparison

Method	Training Time (hrs)	Communication Cost	Inference Latency (ms/img)
Centralized	3.2	None	12
Federated Learning	5.8	Moderate–High	14
Differential Privacy	4.1	None	13
Homomorphic Encryption	3.5	Low	48

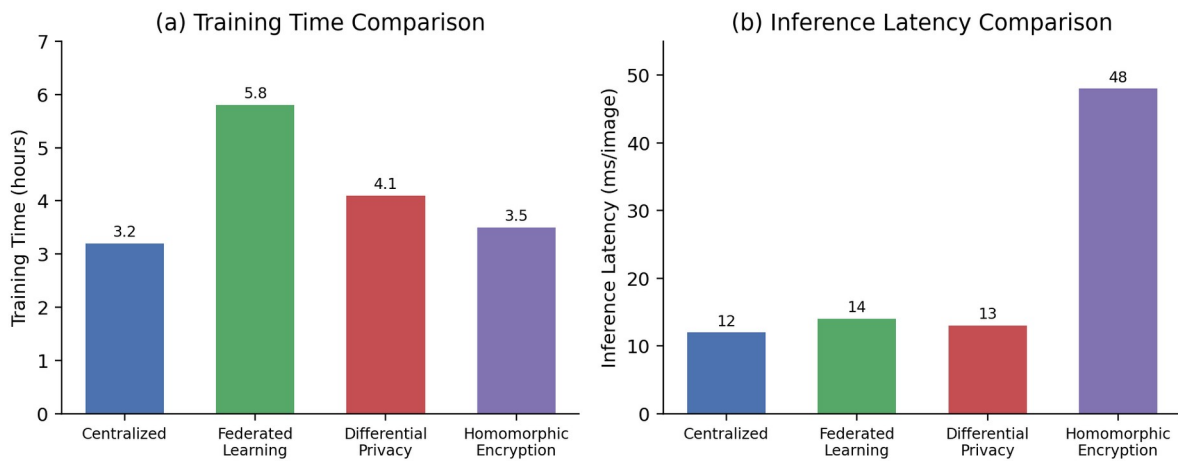


Fig. 4. Training Time and Inference Latency Comparison Across Privacy-Preserving Paradigms.

The two costliest paradigms are expensive in different currencies. FL nearly doubles training time (5.8 vs. 3.2 hours) because of its 100 communication rounds — a training-time cost with no effect on deployed inference speed. HE does the opposite: training stays cheap (3.5 hours), but every single inference call pays a 4× latency tax. For a hospital, that distinction matters: FL's cost is a one-time training expense, HE's is a recurring per-scan cost that scales with patient volume.

6.0 Discussion

None of the three paradigms is a strict upgrade over the others — each wins on a different axis. Federated Learning gets closest to centralized accuracy (92.4% vs. 93.1%) while keeping raw scans on-site, which eliminates the most obvious exposure risk outright. It isn't flawless: gradient updates can still leak information without secure aggregation in place, and the 5.8-hour training time reflects a real communication bottleneck that would only get worse on a slower hospital network.

Differential Privacy is the only paradigm here with a formal, tunable privacy guarantee, and it delivers measurably better resistance to membership-inference attacks — but the accuracy cost is the steepest of the three, and it gets worse fast as ϵ tightens. Choosing ϵ is therefore not a technical afterthought; it is close to the central design decision for any DP deployment, since the difference between $\epsilon = 1.0$ and $\epsilon = 0.5$ alone was 1.8 accuracy points in this study.

Homomorphic Encryption gave the strongest confidentiality guarantee of the three, encrypting data through the entire inference computation, but at a cost that is hard to ignore clinically: a 4× latency increase is a real problem in emergency settings where a diagnosis needs to come back in seconds, not tens of milliseconds turning into hundreds.

Matched against deployment context, the pattern is fairly clean: FL suits large hospital networks that can tolerate the communication overhead in exchange for near-centralized accuracy; DP fits institutions under strict regulatory pressure that can absorb some accuracy loss; and HE fits high-security cloud deployments where confidentiality outweighs speed. Combining FL with DP — local training plus noised updates — is a plausible next step for institutions that need both properties at once, though this study did not test that combination directly.

7.0 Security Threat Modeling

Four attack types matter most for medical AI deployments: model inversion, membership inference, data poisoning, and adversarial perturbation — and the three paradigms don't defend against them equally.

Model inversion tries to reconstruct training images from a trained model's outputs. Centralized models are the most exposed; FL reduces exposure by keeping data local but can still leak through gradients absent secure aggregation; DP limits any single training example's influence directly; and HE, by keeping data encrypted through computation, offers the strongest protection of the four against this specific attack.

Membership inference — determining whether a specific patient's scan was used in training — follows a similar pattern: centralized models are most vulnerable, DP meaningfully lowers attack success, FL offers partial protection contingent on secure aggregation, and HE blocks the direct data access this attack typically relies on.

Data poisoning is a different threat entirely — it targets training integrity, not confidentiality. It is FL's clearest weak spot: a compromised client can inject corrupted gradients into the aggregation round, and without anomaly detection on incoming updates, that corruption can spread into the global model. Neither DP nor HE address this directly; DP's noise doesn't detect malicious updates, and HE's encryption protects confidentiality but says nothing about integrity.

Adversarial perturbation — small, deliberately crafted input changes that flip a model's prediction — sits outside what any of these three paradigms were designed to solve. All three need adversarial training or input validation layered on top if this is a real concern for the deployment.

8.0 Regulatory Compliance Perspective

HIPAA's core requirement — protect patient health information — maps onto these paradigms in different ways. FL satisfies it structurally, since raw data never crosses institutional boundaries in the first place. DP satisfies it statistically, by bounding how much any single patient's data can influence — and therefore be inferred from — the trained model. HE satisfies it cryptographically, keeping data encrypted through the full computation.

GDPR's emphasis on data minimization and cross-border safeguards lines up the same way: FL keeps data local by construction, DP offers a quantifiable privacy budget (ϵ) that can be documented for compliance review, and HE secures cross-border cloud processing when multinational data-sharing agreements are involved. One practical detail specific to medical imaging: DICOM files carry metadata that needs its own protection — TLS/SSL in transit, anonymization at rest — independent of whichever training paradigm is used. FL sidesteps most inter-hospital DICOM transfer altogether, while HE is the better fit for securing DICOM data that does need to move through a cloud inference pipeline.

9.0 Conclusion

Run head-to-head on the same datasets, architecture, and metrics, Federated Learning, Differential Privacy, and Homomorphic Encryption each solve a different part of the privacy problem, and none of them solves all of it. FL came within 0.7 accuracy points of centralized training while eliminating raw data transfer. DP delivered the only formally provable privacy guarantee of the three, at a real and ϵ -dependent accuracy cost. HE delivered the strongest confidentiality guarantee outright, at a roughly four-fold inference latency cost. Deployment choice, on this evidence, should follow institutional constraints rather than a single 'best' paradigm.

This work has real limits. All three paradigms were evaluated in simulation — FL clients, DP noise, and HE encryption were all run on one workstation rather than across genuinely distributed hospital infrastructure, and results may shift under real network latency, heterogeneous hardware, and true non-IID data across sites. Every number reported here also comes from a single training run rather than an average over repeated runs, so reported accuracies should be read as point estimates rather than statistically confirmed differences. Follow-up work should validate this comparison on a live multi-institutional deployment, test a combined FL+DP configuration directly, and evaluate hardware-accelerated HE implementations aimed at closing the inference-latency gap.

References

- [1] McMahan, H.B., Moore, E., Ramage, D., Hampson, S., y Arcas, B.A.: Communication-efficient learning of deep networks from decentralized data. In: Proc. AISTATS (2017)

- [2] Bonawitz, K., Ivanov, V., Kreuter, B., Marcedone, A., McMahan, H.B., et al.: Practical secure aggregation for privacy-preserving machine learning. In: ACM CCS (2017)
- [3] Kairouz, P., McMahan, H.B., et al.: Advances and open problems in federated learning. *Found. Trends Mach. Learn.* 14(1–2), 1–210 (2021)
- [4] Sheller, M.J., Edwards, B., Reina, G.A., Martin, J., Pati, S., et al.: Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Sci. Rep.* (2020)
- [5] Dayan, I., Roth, H.R., Zhong, A., Harouni, A., Gentili, A., Abidin, A.Z., et al.: Federated learning for predicting clinical outcomes in patients with COVID-19. *Nat. Med.* 27, 1735–1743 (2021)
- [6] Abadi, M., Chu, A., Goodfellow, I., McMahan, H.B., Mironov, I., Talwar, K., Zhang, L.: Deep learning with differential privacy. In: ACM CCS (2016)
- [7] Ziller, A., et al.: Medical imaging deep learning with differential privacy. *Sci. Rep.* (2021)
- [8] Bu, Z., Dong, J., Su, W., et al.: Deep learning with Gaussian/f-differential privacy — theory and practice (2020)
- [9] Papernot, N., Abadi, M., Erlingsson, Ú., Goodfellow, I., Talwar, K.: Semi-supervised knowledge transfer for deep learning from private training data. In: ICLR/arXiv (2016)
- [10] Hu, H., Salicic, Z., Sun, L., Dobbie, G., Yu, P.S., Zhang, X.: Membership inference attacks on machine learning: a survey. *ACM Comput. Surv.* (2022)
- [11] Wood, A., Najarian, K., Kahrobaei, D.: Homomorphic encryption for machine learning in medicine and bioinformatics. *ACM Comput. Surv.* 53(4) (2020)
- [12] Dowlin, N., Gilad-Bachrach, R., Laine, K., Lauter, K., Naehrig, M., Wernsing, J.: CryptoNets: applying neural networks to encrypted data with high throughput and accuracy. In: ICML (2016)
- [13] Hu, X., Tran, A.T., et al.: A comprehensive survey and taxonomy on privacy-preserving deep learning. *Neurocomputing* (2024)
- [14] He, Z., et al.: SecureBadger: a homomorphic encryption-based framework for secure medical inference. *ScienceDirect* (2025)
- [15] Li, T., Sahu, A.K., Talwalkar, A., Smith, V.: Federated learning: challenges, methods, and future directions. *IEEE Signal Process. Mag.* 37(3), 50–60 (2020)
- [16] Rieke, N., Hancox, J., Li, W., Milletari, F., Roth, H.R., et al.: The future of digital health with federated learning. *NPJ Digit. Med.* 3, 119 (2020)
- [17] Kaissis, G.A., Makowski, M.R., Rückert, D., Braren, R.F.: Secure, privacy-preserving and federated machine learning in medical imaging. *Nat. Mach. Intell.* 2, 305–311 (2020)
- [18] Brisimi, T.S., Chen, R., Mela, T., Olshevsky, A., Paschalidis, I.C., Shi, W.: Federated learning of predictive models from electronic health records. *Int. J. Med. Inform.* 112, 59–67 (2018)
- [19] Truex, S., Liu, L., Chow, K., Gursoy, M.E., Wei, W.: LDP-Fed: federated learning with local differential privacy. In: Proc. 3rd ACM Int'l Workshop on Edge Systems, Analytics and Networking (EdgeSys) (2020)
- [20] Geyer, R.C., Klein, T., Nabi, M.: Differentially private federated learning: a client-level perspective. *arXiv* (2017)
- [21] Dwork, C., Roth, A.: The algorithmic foundations of differential privacy. *Found. Trends Theor. Comput. Sci.* 9(3–4), 211–407 (2014)
- [22] Mironov, I.: Rényi differential privacy. In: IEEE CSF Symposium (2017)
- [23] Acar, A., Aksu, H., Uluagac, A.S., Conti, M.: A survey on homomorphic encryption schemes: theory and implementation. *ACM Comput. Surv.* 51(4) (2018)
- [24] Chillotti, I., Gama, N., Georgieva, M., Izabachène, M.: Faster fully homomorphic encryption: bootstrapping in less than 0.1 seconds. In: ASIACRYPT (2016)
- [25] Yang, Q., Liu, Y., Chen, T., Tong, Y.: Federated machine learning: concept and applications. *ACM TIST* 10(2) (2019)
- [26] Konečný, J., McMahan, H.B., Yu, F.X., Richtárik, P., Suresh, A.T., Bacon, D.: Federated learning: strategies for improving communication efficiency. *arXiv* (2016)
- [27] Shokri, R., Stronati, M., Song, C., Shmatikov, V.: Membership inference attacks against machine learning models. In: IEEE S&P (2017)
- [28] Goodfellow, I., Bengio, Y., Courville, A.: Deep Learning. MIT Press (2016)
- [29] Litjens, G., Kooi, T., Bejnordi, B.E., Setio, A.A.A., Ciompi, F., et al.: A survey on deep learning in medical image analysis. *Med. Image Anal.* 42, 60–88 (2017)
- [30] Esteva, A., Kuprel, B., Novoa, R.A., Ko, J., Swetter, S.M., Blau, H.M., Thrun, S.: Dermatologist-level classification of skin cancer with deep neural networks. *Nature* (2017)

© 2026 The Author(s). Published by WhiteField Academic Press. This is an open access article distributed under the terms of the Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. Authors retain copyright.



INTERNATIONAL JOURNAL OF ADVANCED COMPUTING AND MACHINE INTELLIGENCE

*Peer-Reviewed · Open Access · Published by WhiteField
Academic Press
Noida, India*



Scan to visit

www.ijacmi.org
editor@ijacmi.org
submissions@ijacmi.org

Visit our website — www.ijacmi.org

